



Ex 5.22 F 可微, ψ 严格凸 (补充, 补充)

梯度下降实际上也是一种距离极小化

$$Q(\eta) = F(\theta) + g^T(\eta - \theta) + \frac{1}{2\gamma} \|\eta - \theta\|_2^2, \eta =$$

所以梯度下降可以转换为一个等价的极小化问题

$$Q'(\eta) = g + \frac{1}{\gamma}(\eta - \theta) \quad \eta = \theta - \gamma g$$

$$\nabla^2 Q(\eta) = \frac{1}{\gamma} I \succ 0 \quad \text{Hesse 阵正定}$$

那么如果换成别的近似可以吗?

$$D_\psi(\eta, \theta) = \psi(\eta) - \psi(\theta) - \psi'(\theta)^T(\eta - \theta) \text{ 为 Bregman 散度}$$

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Coding KNN Ch2 - 1 部分, 整理代码仓库 Deep-learning 整理 Mat

5.20.

$$5.22 \quad D_\psi \geq 0 \quad D_\psi = 0 \Leftrightarrow \eta = \theta$$

$$\text{已知 } \psi(\eta) \quad \eta = \theta + (\eta - \theta) = \theta + tV$$

$$\psi(\theta + tV) - \psi(\theta) - t\psi'(\theta)^T V \quad \lim_{t \rightarrow 0} \frac{\psi(\theta + tV) - \psi(\theta)}{t} \leq \frac{t\psi'(\theta)^T V}{t} = \psi'(\theta)^T V$$

$$\forall t \in (0, 1) \quad \psi(\theta + tV) - \psi(\theta) \leq t(\psi(\theta + V) - \psi(\theta)) = \frac{t\psi(\theta + V) + (1-t)\psi(\theta) - \psi(\theta)}{t} \geq \frac{\psi(\theta + tV) - \psi(\theta)}{t}$$

$$\text{令 } t \rightarrow 0 \quad \psi(\theta + V) - \psi(\theta) \geq \lim_{t \rightarrow 0} \frac{\psi(\theta + tV) - \psi(\theta)}{t} = \psi'(\theta)^T V$$

$$D_\psi \geq 0 \quad \text{而 } \psi(\theta + V) - \psi(\theta) = \psi'(\theta)^T V, \text{ 只有 } \psi(\theta + V) - \psi(\theta) = \lim_{t \rightarrow 0} \frac{\psi(\theta + tV) - \psi(\theta)}{t}$$

此时 $\psi(\theta + V) = \psi(\theta) \quad \eta = \theta$ 才行, 因为 ψ 严格凸



(3) 考虑极小化 $F(\theta) + F'(\theta)^T(\eta - \theta) + \frac{1}{\gamma} D_{\psi}(\eta, \theta)$ 的 η

$$F'(\theta) + \frac{1}{\gamma} D'_{\psi}(\eta, \theta) = 0 \quad (\text{求导})$$

$$F'(\theta) + \frac{1}{\gamma} (\psi'(\eta) - \psi'(\theta)) = 0$$

$$\psi'(\eta) = \psi'(\theta) - \frac{1}{\gamma} F'(\theta)$$

而且, ψ 只定义在开的凸集 K 上, ψ' 是从 K 到 $\mathbb{R}^d$ 上双射, 同样的结论成立

$$\eta \in \operatorname{argmin}_{\xi \in K} (F(\theta) + F'(\theta)^T(\xi - \theta) + \frac{1}{\gamma} D_{\psi}(\xi, \theta))$$

$$(4) \quad F(\theta) + F'(\theta)^T(\xi - \theta) + \frac{1}{\gamma} (\psi(\eta) - \psi(\theta) - \psi'(\theta)^T(\xi - \eta))$$

$$\text{实际极小化 } \frac{1}{\gamma} (\psi(\xi) - \psi'(\theta)^T \xi) + F'(\theta)^T \xi$$

$$\operatorname{argmin}_{\xi} \frac{1}{\gamma} (\psi(\xi) - (\psi'(\theta) - \gamma F'(\theta))^T \xi)$$

$$\text{先证 } \exists \eta \in K, \psi'(\eta) = \psi'(\theta) - \gamma F'(\theta) = u$$

$$\text{由于 } u = \psi'(\theta) - \gamma F'(\theta) \in \mathbb{R}^d, \psi' \text{ 是从 } K \rightarrow \mathbb{R}^d$$

由于 K 是开集, $\eta \in K$, η 为 K 上内点, 可以讨论导数, $\nabla Q_0 = 0$

由于 Q_0 在凸集上 K 严格凸, 驻点即为全局极小

$$(4) \quad \psi(\theta) = \sum \theta_i \log \theta_i \quad \psi'(\theta)_i = (1 + \log \theta_1, 1 + \log \theta_2, \dots, 1 + \log \theta_n)$$

$$(1 + \log \eta_1, 1 + \log \eta_2, \dots, 1 + \log \eta_n) = (1 + \log \theta_1) - \frac{1}{\gamma} \gamma F'(\theta)$$

$$\eta_i = \theta_i \exp(-\gamma F'_i(\theta))$$

$$\eta = \theta \otimes \exp(-\gamma F'(\theta)) \quad \otimes \text{逐元素相乘}$$

$$\theta_{t+1} = \theta_t \otimes \exp(-\gamma F'(\theta_t))$$



Ex 5.23. 考虑 5.21 同轴算法的正交映射

$$\Theta \in K, \theta^* = \arg \min_{\theta \in K} F(\theta) \quad \Theta_{t+1} = \Pi(\Theta_t - \gamma_t F'(\Theta_t)) \quad \gamma_t = \frac{D}{B\sqrt{t}}$$

$$\text{由 Ex 5.21} \quad \|\Theta_t - \theta^*\|_2^2 \leq \|\Theta_{t-1} - \theta^*\|_2^2 - 2\gamma_t F'(\Theta_{t-1})^T (\Theta_{t-1} - \theta^*) + \gamma_t^2 \|F'(\Theta_{t-1})\|_2^2$$

$$\text{已知 } (F(\Theta_t) - F(\theta^*)) \leq F'(\Theta_{t-1})^T (\Theta_{t-1} - \theta^*) \leq \frac{\|\Theta_{t-1} - \theta^*\|_2^2 - \|\Theta_t - \theta^*\|_2^2}{\gamma_t} + \gamma_t \|F'(\Theta_{t-1})\|_2^2$$

$$2 \left(\sum_{s=1}^t F(\Theta_s) - t F(\theta^*) \right) \leq \sum_{s=1}^t \frac{\|\Theta_{s-1} - \theta^*\|_2^2 - \|\Theta_s - \theta^*\|_2^2}{\gamma_s} + \sum_{s=1}^t \gamma_s B^2 \leq \text{Sum} + 2BD\sqrt{t}$$

$$\sum_{s=1}^t \gamma_s B^2 = \sum_{s=1}^t BD \frac{1}{\sqrt{s}} \leq 2BD\sqrt{t}$$

$$\text{关键在于处理 } \sum_{s=1}^t \frac{1}{\gamma_s} (\|\Theta_{s-1} - \theta^*\|_2^2 - \|\Theta_s - \theta^*\|_2^2) = \frac{1}{\gamma_1} \|\Theta_0 - \theta^*\|_2^2 + \sum_{s=1}^{t-1} \left(\frac{1}{\gamma_{s+1}} - \frac{1}{\gamma_s} \right) \|\Theta_s - \theta^*\|_2^2 - \frac{1}{\gamma_t} \|\Theta_t - \theta^*\|_2^2$$

$$\text{第一项 } D^2 \cdot \frac{B}{D} \cdot 1$$

$$\text{第二项求和 } \sum_{s=1}^{t-1} D^2 \cdot \frac{B}{D} (\sqrt{s+1} - \sqrt{s})$$

$$\text{第三项 } -\frac{1}{\gamma_t} \|\Theta_t - \theta^*\|_2^2 \leq -\sqrt{t} BD$$

$$\text{最终: } BD\sqrt{t}$$

$$\text{综上 } 2 \sum_{s=1}^t (F(\Theta_s) - F(\theta^*)) \leq 3BD\sqrt{t}$$

$$2t(F(\bar{\Theta}_t) - F(\theta^*)) \leq 3BD\sqrt{t} \quad F(\bar{\Theta}_t) - F(\theta^*) \leq \frac{3BD}{2\sqrt{t}}$$

回到算法, 之前选取 $\gamma_s = \frac{D}{B\sqrt{s}}$, 可以选取固定步长吗?

Actually, 可以, 当 $\gamma = \frac{D}{B\sqrt{T}}$, 在 T 之前停止都有保证

$$\text{此时 } F(\Theta_{t+1}) - F(\theta^*) \leq \frac{\|\Theta_t - \theta^*\|_2^2 - \|\Theta_{t+1} - \theta^*\|_2^2}{2\gamma} + \frac{\gamma B^2}{2}$$

$$\sum_{t=1}^T (F(\Theta_t) - F(\theta^*)) \leq \frac{\|\Theta_0 - \theta^*\|_2^2 - \|\Theta_T - \theta^*\|_2^2}{2\gamma} + \frac{T\gamma B^2}{2}$$

$$F(\bar{\Theta}_T) - F(\theta^*) \leq \frac{D^2}{2\gamma T} + \frac{\gamma B^2}{2} \quad \text{这样就只有 } \frac{1}{\sqrt{T}}, \text{ 没有 } \log t$$

$$\text{一般步长 } \sum \gamma_i^2 \leq 1 + \log t \quad \gamma_i = \frac{1}{\sqrt{i}} \text{ 时引入 } \log t$$

但此时不是 anytime, 之前跑到哪都有 $\frac{1}{\sqrt{T}}$ 的限制, 但固定步长要跑满 T 步

因为之前若 $T = 10^4$, $t = 100$, 之前走得太慢



武汉大学

WUHAN UNIVERSITY

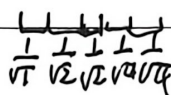
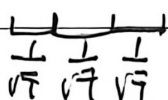
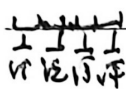
double trick

若未知最终运行时间, 那么把运行过程分成 $T_k = 2^k$

对于第 k 个阶段 $\gamma_k = \frac{D}{B\sqrt{T_k}} = \frac{D}{B\sqrt{2^k}}$

一般

若



验证 double trick 下的收敛速度 Ex 5.24

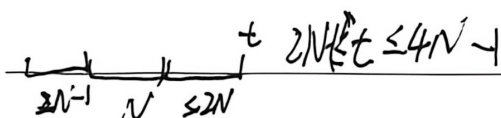
$$\min_{0 \leq s \leq t} E[F(\theta_s)] - F(\theta^*) \leq C \frac{DB}{\sqrt{t}}$$

首先考虑一个完整的区间.

$$\frac{1}{N} E[F(\theta_j)] - F(\theta^*) \leq \frac{DB}{\sqrt{N}}$$

$$\min E[F(\theta_j)] - F(\theta^*) \leq \frac{DB}{\sqrt{N}}$$

其次 $N \geq \frac{1}{4}t$, 最大完整区间至少整个 t 的 $\frac{1}{4}$



$$\min_{N \leq s \leq 2N} E[F(\theta_s)] - F(\theta^*) \leq \min_{N \leq s \leq 2N} E[F(\theta_s)] - F(\theta^*) \leq \frac{DB}{\sqrt{N}} \leq \frac{2DB}{\sqrt{t}}$$

Ex 5.25 计算 B, D, γ , 对于

(1) 正则化的 logistic 模型

$$F(\theta) = \frac{1}{n} \sum_{i=1}^n \log(1 + e^{-y_i \theta^T x_i}) + \frac{\lambda}{2} \|\theta\|_2^2$$

$$B = \frac{1}{n} \sum_{i=1}^n \frac{e^{-y_i \theta^T x_i}}{1 + e^{-y_i \theta^T x_i}} (-y_i x_i) + \lambda \theta$$

$$B \leq \frac{1}{n} \sum_{i=1}^n \max_i \|x_i\|_2 + \lambda \|\theta\|_2 \leq R + \lambda \|\theta\|_2$$

$$F(\theta^*) \leq F(0) = \log 2$$

$$\frac{\lambda}{2} \|\theta\|_2^2 \leq \log 2 \quad \|\theta\|_2 \leq \sqrt{\frac{2 \log 2}{\lambda}}$$



$$D = \|\theta_0 - \theta^*\|_2 = \|\theta^*\|_2 \leq \sqrt{\frac{2(\log 2)}{\lambda}} \quad B \leq R + \frac{1}{\lambda} \sqrt{\frac{2(\log 2)}{\lambda}}$$

或者利用 $\nabla F(\theta^*)$

$$\lambda \theta^* + \frac{1}{n} \sum_{i=1}^n \frac{e^{-y_i \theta^{*T} x_i}}{1 + e^{-y_i \theta^{*T} x_i}} (-y_i x_i) = 0$$

$$\|\lambda \theta^*\|_2 \leq R$$

$$B \leq 2R \quad D \leq \frac{R}{\lambda} \quad D = \|\theta^* - \theta_0\| \leq \frac{R}{\lambda}$$

支持向量机 $l_1 + l_2$ 正则

$$\ell(\theta) = \frac{1}{n} \sum_{i=1}^n (1 - y_i \theta^T x_i) + \lambda \|\theta\|_2^2$$

$$\partial \ell(\theta) = \begin{cases} -y_i \frac{1}{n} \sum_{i=1}^n y_i x_i & y_i \theta^T x_i < 1 \\ -\frac{1}{n} & \end{cases}$$

$$\partial \ell_i(\theta) = \begin{cases} -y_i x_i & y_i \theta^T x_i < 1 \\ -2y_i x_i & y_i \theta^T x_i = 1, \alpha \in [0, 1] \\ 0 & y_i \theta^T x_i > 1 \end{cases}$$

因此 $B = \|\partial \ell(\theta)\| \leq R + \lambda \|\theta\|$, $R = \|x_i\|_2 \leq R$, 特征元最大

$$\ell(\theta^*) = \lambda \|\theta^*\|_2^2 \leq \ell(\theta^*) \leq \ell(0) = 1 \quad (\text{值估计}) \quad (R + \frac{1}{\lambda}) (R + \frac{1}{\lambda}) \frac{1}{\lambda}$$

$$\|\lambda \theta\| = \frac{1}{n} \sum \partial \ell_i(\theta) \leq R \quad (\text{梯度估计})$$

$$B \leq 2R \quad \|\theta^*\| \leq \frac{R}{\lambda}$$

$$BD \leq \frac{2R^2}{\lambda}$$

在实际计算 $B = GR$ G : ℓ 的 Lipschitz 常数 R 为特征的上界

D 为参数矩域 选取 $\gamma_t = \frac{BD}{\sqrt{t}}$

总误差优化误差 $\frac{GRD}{\sqrt{t}}$ $t \sim n$, 两误差只差常数